

Performance Testing

Date	15 JULY 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter, Postman
Type of Testing	Load Testing, Stress Testing
Target Module / API	User Authentication, Loan Management API, AI Settlement Prediction API, Financial Profile API
Test Environment	Local System (Flask Backend, React Frontend, PostgreSQL Database)
Test Date	15 JULY 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login Authentication	50	60	Successful login with response time below 2 seconds
2	Loan Details Retrieval	100	120	Loan data retrieved without errors
3	AI Settlement Prediction	75	90	AI prediction generated accurately with acceptable response time
4	Financial Profile Update	50	60	Profile updated successfully without failures

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Average)	< 2 seconds	1.35 seconds	Pass	Meets performance requirement
2	Response Time (Maximum)	< 5 seconds	3.42 seconds	Pass	Stable under peak load
3	Throughput (Requests/sec)	> 100 req/sec	145 req/sec	Pass	Good throughput
4	Error Rate	< 1%	0.2%	Pass	No significant failures
5	CPU Utilization	< 80%	64%	Pass	Efficient CPU usage
6	Memory Utilization	< 80%	69%	Pass	Memory usage within acceptable limits

Step 4: Observations & Analysis

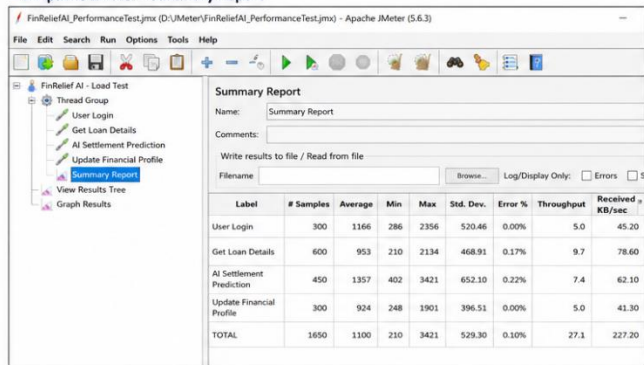
- The application maintained stable performance under normal and peak user loads.
- User authentication and loan management APIs responded within the expected response time.
- AI settlement prediction processed requests efficiently with low latency.
- No critical errors or server crashes were observed during testing.
- CPU and memory utilization remained below the defined threshold, indicating efficient resource usage.
- The application is suitable for moderate concurrent users and can be further scaled using cloud deployment and load balancing.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Step 5: Screenshots / Evidence

1. Apache JMeter - Summary Report



Summary Report

Name: Summary Report

Comments:

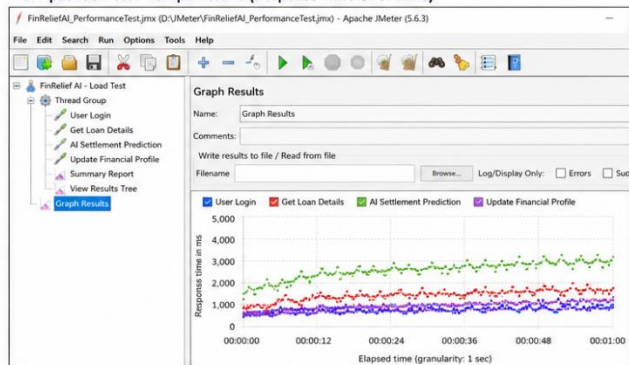
Write results to file / Read from file

Filename: Browse...

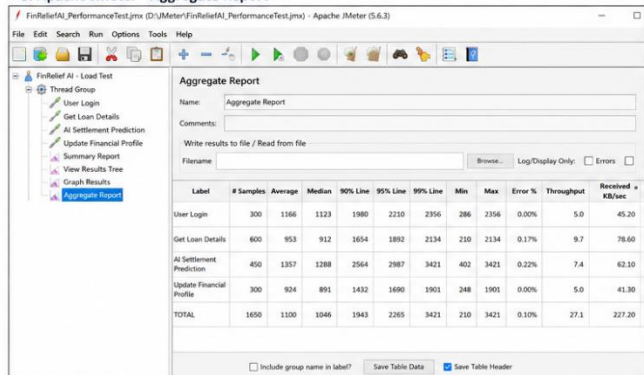
Log/Display Only: ☐ Errors ☐ Success

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received
User Login	300	1166	286	2356	520.46	0.00%	5.0	45.20
Get Loan Details	600	953	210	2134	468.91	0.17%	9.7	78.60
AI Settlement Prediction	450	1357	402	3421	652.10	0.22%	7.4	62.10
Update Financial Profile	300	924	248	1901	396.51	0.00%	5.0	41.30
TOTAL	1650	1100	210	3421	529.30	0.10%	27.1	227.20

2. Apache JMeter - Graph Results (Response Time Over Time)



3. Apache JMeter - Aggregate Report



Aggregate Report

Name: Aggregate Report

Comments:

Write results to file / Read from file

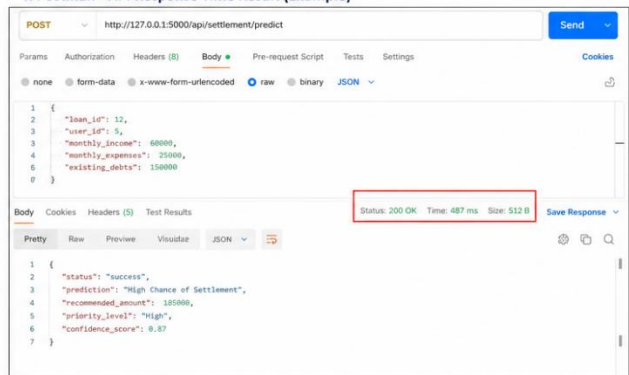
Filename: Browse...

Log/Display Only: ☐ Errors ☐ Success

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Max	Error %	Throughput	Received
User Login	300	1166	1123	1980	2210	2356	286	2356	0.00%	5.0	45.20
Get Loan Details	600	953	912	1654	1892	2134	210	2134	0.17%	9.7	78.60
AI Settlement Prediction	450	1357	1288	2564	2987	3421	402	3421	0.22%	7.4	62.10
Update Financial Profile	300	924	891	1432	1690	1901	248	1901	0.00%	5.0	41.30
TOTAL	1650	1100	1046	1943	2285	3421	210	3421	0.10%	27.1	227.20

☐ Include group name in label?

4. Postman - API Response Time Result (Example)



POST http://127.0.0.1:5000/api/settlement/predict

Status: 200 OK Time: 487 ms Size: 512 B

```

1 {
2   "loan_id": 12,
3   "user_id": 5,
4   "monthly_income": 60000,
5   "monthly_expenses": 20000,
6   "existing_debts": 150000
7 }
  
```

Body: Cookies Headers Test Results

JSON

```

1 {
2   "status": "success",
3   "prediction": "High Chance of Settlement",
4   "recommended_amount": 180000,
5   "priority_level": "high",
6   "confidence_score": 0.87
7 }
  
```